

FIKA

Modular cargo bike platform for urban work and family life

A design brief for a configurable mobility ecosystem built around one durable chassis and a family of exchangeable modules.



One platform. Many urban lives.

FIKA addresses the gap between a bicycle and a light commercial vehicle. It gives urban households and small operators one compact mobility base that can change purpose as often as their day does.



CORE VALUE

Modularity gives one vehicle the flexibility of many.

PART A

WHY

Urban friction and market gap

PART B

WHAT

Platform, modules and visual DNA

PART C

HOW

Engineering, lifecycle and rollout

The gap between a van and a bicycle

Urban operators lose time to congestion, parking and vehicle overhead. Families face the same friction when one trip combines children, groceries and bulky everyday loads.

- 01 Vans carry cost and space even when the load is small.
- 02 Fixed cargo bikes solve one task but adapt poorly to the next.
- 03 Proprietary bodies and electronics raise repair and replacement costs.

Opportunity: separate the durable vehicle platform from the task-specific body, then let both evolve on different replacement cycles.



Three routines, one mobility platform

FIKA serves business users who need productive uptime and households that need flexibility without a car. Each persona values a different module, but every one depends on stable handling, quick changeover and simple service.



B2B / COURIER

Delivery rider

A rider carries parcels through dense streets, needs a predictable range and wants a lockable box that reports temperature, door status and payload.



B2B / MOBILE RETAIL

Coffee operator

A small business moves between commuter peaks and events. The service module must open quickly, carry power safely and present a recognizable storefront.



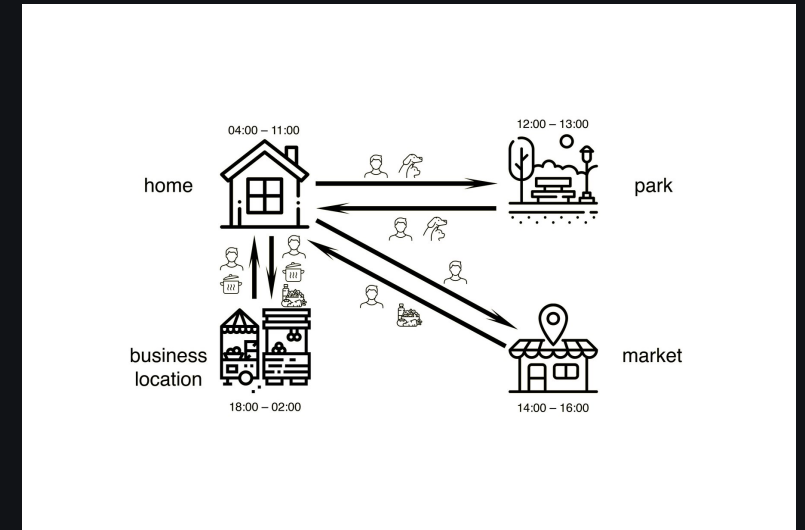
B2C / URBAN FAMILY

Car-light household

A parent handles school drop-off, groceries and a weekend outing. Child safety and low-speed balance matter as much as cargo capacity.

The module changes when the day changes

A household can begin with a child seat, exchange it for an open cargo deck after school, then collect a temperature-controlled box for an evening delivery shift. The platform retains the rider setup, battery state and navigation profile through every swap.



07:30

CHILD SEAT

School drop-off and safe buckle confirmation

11:30

CARGO TRAY

Groceries, returns and everyday errands

16:00

DELIVERY BOX

Secure parcels and route management

19:30

THERMAL BOX

Food service with temperature monitoring

module

middle cargo rack

tricycle platform

riding platform

sliding table

handlebar

windshield

FIKA USP

Platform first. Modules second.

PLATFORM

A common structural spine carries the rider, battery, steering and three-wheel chassis.

MECHANICAL

Self-locating rails, tapered pins and twin over-center latches support a tool-free swap.

POWER + DATA

A blind-mate IP67 connector carries 48 V power, CAN-FD data and module identification.

SAFETY

Dual latch sensing confirms engagement. The drive interlock blocks assist when a module is unsecured.

TARGET SWAP TIME < 3 MIN

Flat cargo deck to thermal box

The physical interface and HMI operate as one system. Hardware guides the module into position while the display confirms identity, latch state and any module-specific sensors.

01

Park + secure

Automatic hold engages and isolates module power.

02

Release

Two latches open and the protected connector disengages.

03

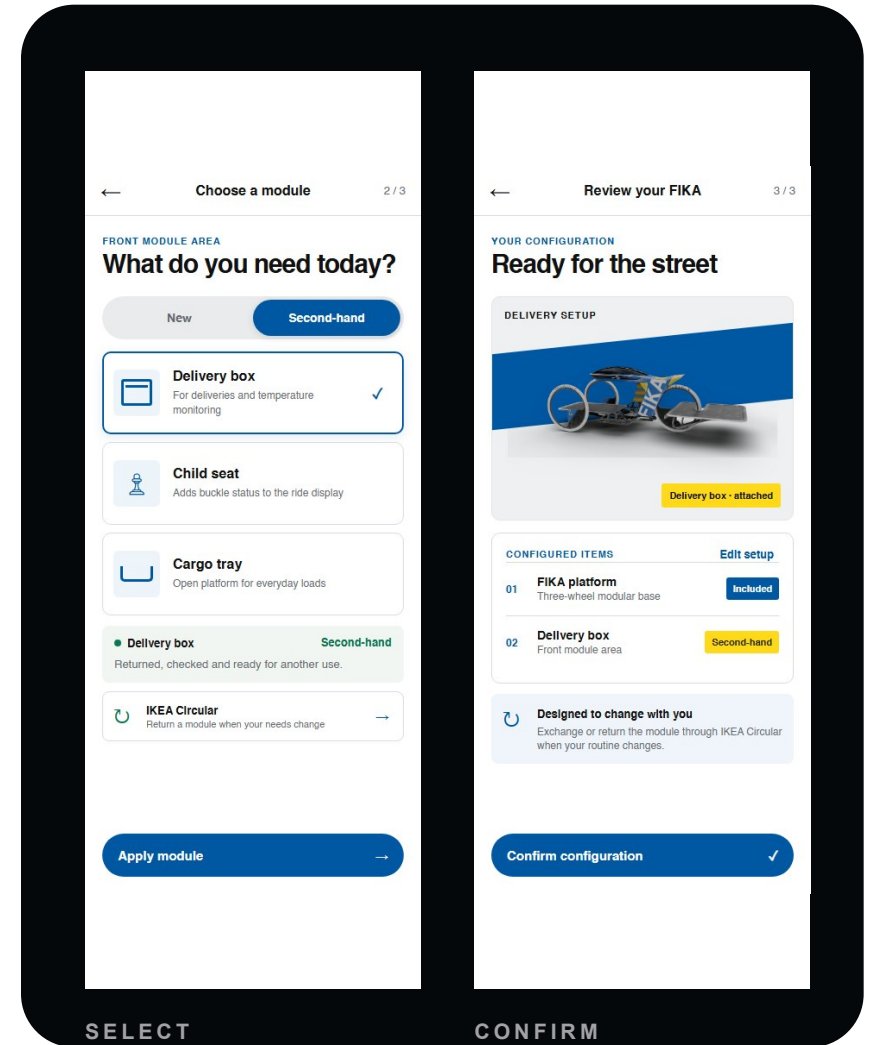
Exchange

The flat deck slides out along low-friction guide rails.

04

Verify

The box locks in; FIKA loads temperature and door status.



A low, stable package for mixed loads

2850 mm

OVERALL LENGTH / SOURCE
DRAWING

951 mm

OVERALL HEIGHT / SOURCE
DRAWING

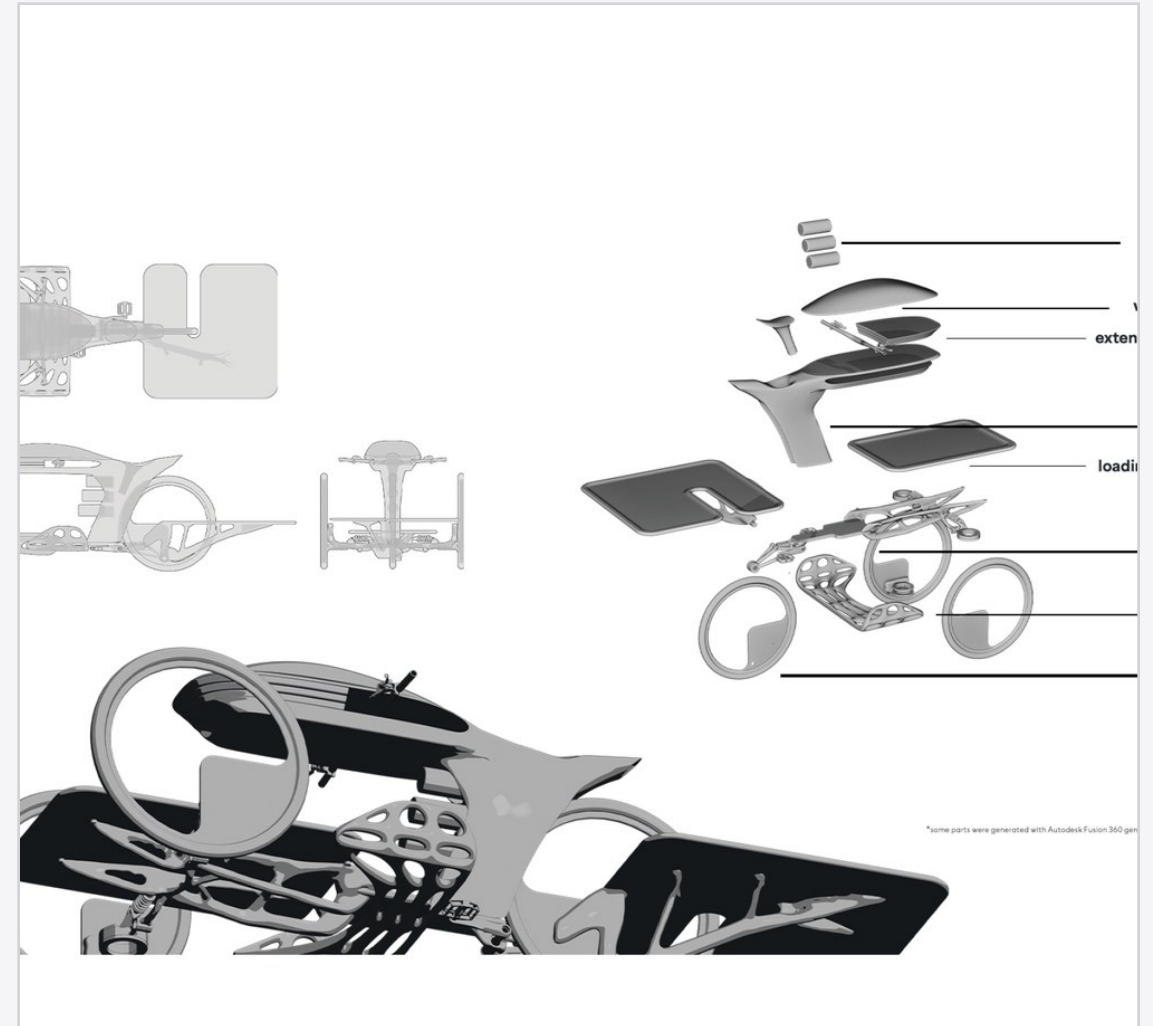
1000 mm

OVERALL WIDTH / SOURCE
DRAWING

~1700 mm

WHEELBASE / DESIGN TARGET

- Loading deck target: approximately 420 mm for lift-in access and a low cargo center of gravity.
- Battery and dense cargo sit low in the central spine; loaded mass stays between the twin-wheel axle and drive wheel.
- The rider triangle targets users from 1550 to 1950 mm through adjustable saddle height and handlebar reach, while preserving steering leverage under load.

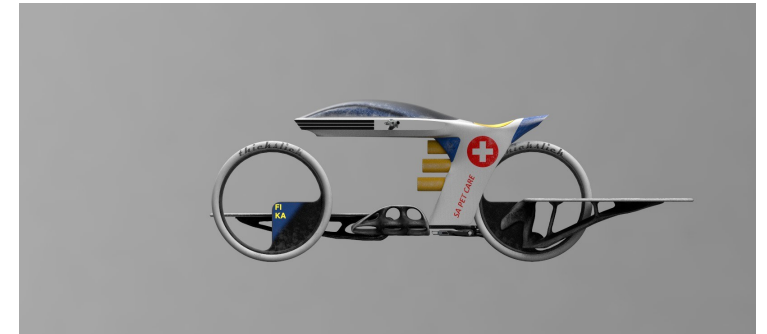
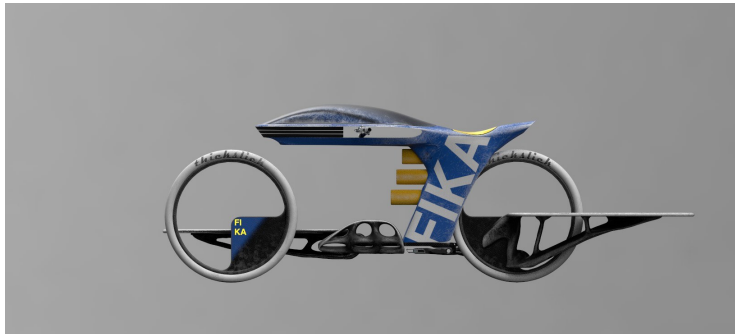


Dimensions in blue are taken from the FIKA package drawing. Yellow values remain design targets pending CAD and physical ergonomic validation.

Functional minimalism with visible utility

VISUAL KEYWORDS

Functional Minimalism / Industrial Clarity / Human-Scale Technology



IKEA BLUE



SAFETY YELLOW



GRAPHITE



SOFT WHITE

RECYCLED ALUMINUM

Primary frame and interface rails; corrosion resistant, repairable and suitable for regional finishing.

RUBBER + TPU

Grip, bumper and sealing surfaces; removable parts simplify service and recovery.

RECYCLED GF NYLON

Module shells and secondary structures; impact tolerant with replaceable wear zones.

LIMITED CFRP

Reserved for weight-critical panels; every use requires a defined disassembly and recovery route.

An architecture sized for city duty

The values below define the first engineering target, not validated production data. Regional power and speed variants should share the same platform, interface and safety logic.



MOTOR	250 W nominal EU / 750 W regional variant
HOT SWAP	Dual bay with protected make-before-break control
PAYLOAD	180 kg module and passenger target
BRAKING	Three-wheel hydraulic disc system with auto hold

BATTERY	48 V, 20 Ah, 0.96 kWh removable pack
RANGE	50-80 km per pack, load and route dependent
ASSIST	25 km/h EU / 32 km/h regional target
SMART	IMU, load sensing, GPS/LTE, lock tracking and module ID

Target specification for concept development. Engineering teams must validate performance, regulatory class, thermal behavior, braking distance and structural safety.

Fewer parts. Longer life.

FIKA lowers manufacturing complexity by keeping the expensive chassis common across uses. Modules contain the task-specific value and can be repaired, refurbished or replaced without retiring the vehicle.

01 COMMON CORE

One chassis, one electrical backbone and one interface across the module family.

02 LOW PART VARIETY

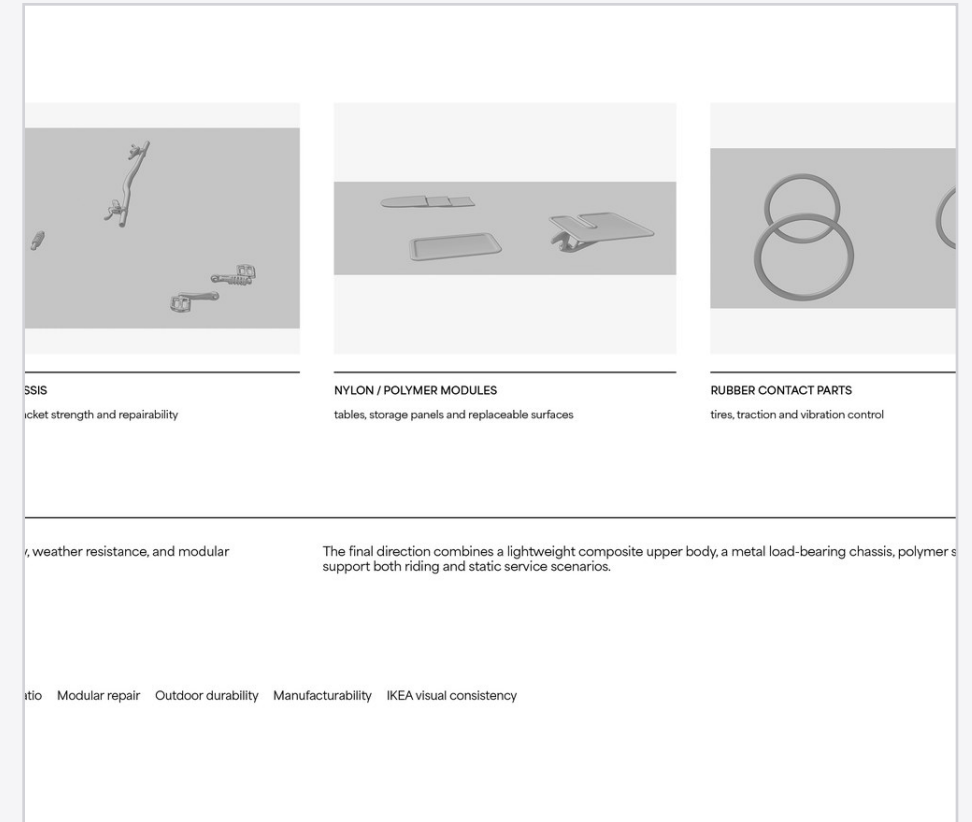
Extrusions, cast nodes and common fasteners replace one-off welded bodies.

03 SERVICE BY ZONE

Wear parts, shells, electronics and latches separate without discarding the structure.

04 TAKE-BACK LOOP

Inspect, refurbish and resell modules; recover aluminum, nylon, rubber and limited CFRP by stream.



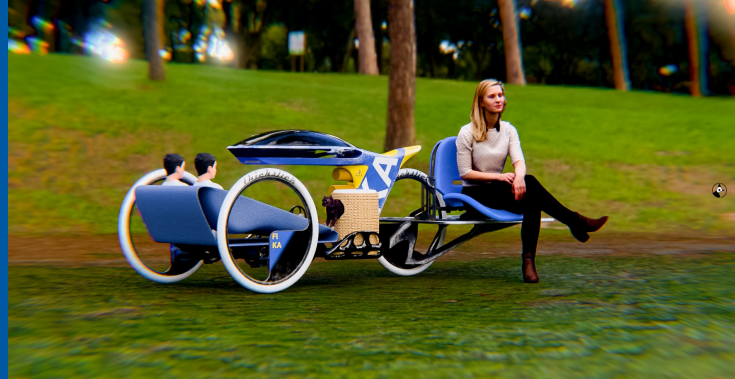
Lifecycle target: chassis 10+ years, modules 5+ years, with replaceable wear components and documented material recovery.

Platform sales, recurring module services



B2B FLEETS

Lease the platform with maintenance, telematics and a module subscription sized to seasonal demand.



B2C OWNERS

Sell the base vehicle directly; add child, cargo and leisure modules through retail and a certified second-life marketplace.



CITY + PARTNERS

Deploy shared fleets through logistics hubs, municipalities and social enterprises. Certify partner-built modules against one interface standard.

0-18 MONTHS

Pilot delivery and family modules

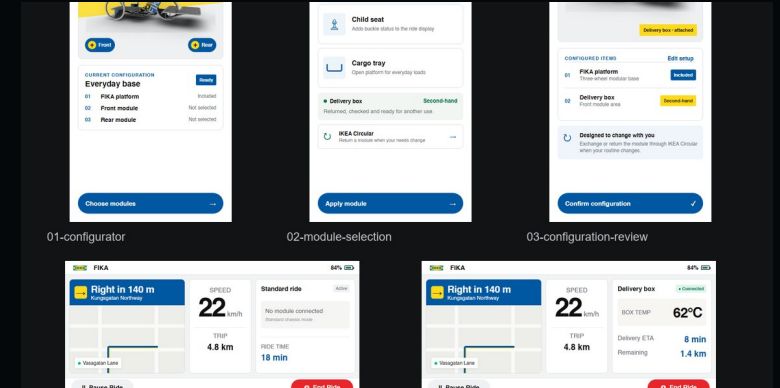
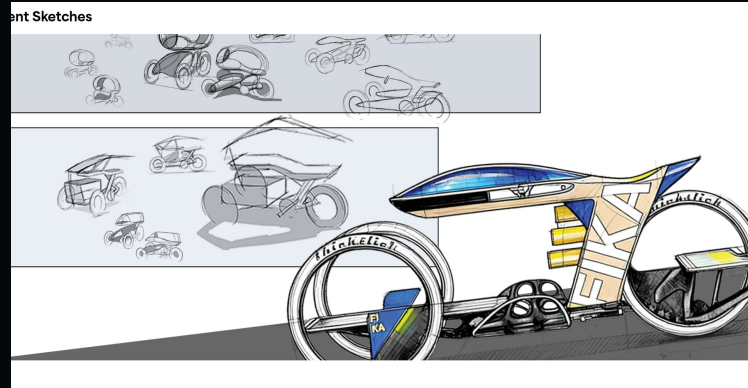
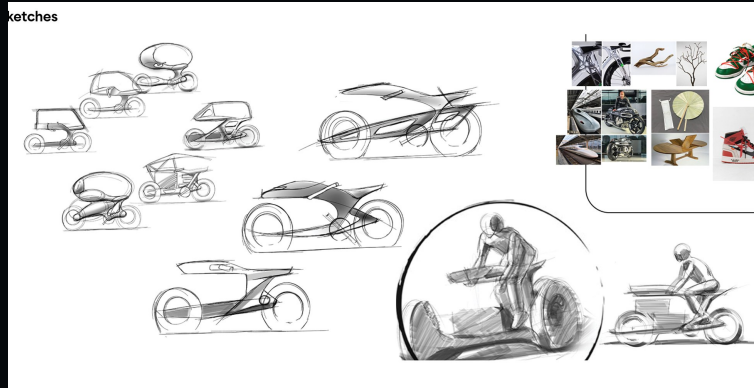
18-36 MONTHS

Add food service, repair and cold-chain modules

36+ MONTHS

Open a certified partner ecosystem

Design package for validation and launch



- Full-size exterior and module renderings
- Orthographic package and loading study
- CMF matrix with material recovery routes
- HMI wireflows and interactive prototype
- B2B/B2C pilot and module roadmap

- Exploded system and interface diagrams
- Rider triangle and loaded handling study
- Engineering target specification
- Manufacturing and service concept
- 60-90 second concept demonstration film

WATCH THE FIKA FILM

All visuals in this brief come from the original FIKA project by Tao Wen / aoT Designs. Dimensions reproduce the project drawing; all values marked target require engineering validation.

aotdesigns.com